

Geometric series



1 20 475

2 a 14.29

b -204.92

3 683

4 $\frac{5461}{64}$

5 17

6 9.143

7 a $\frac{1}{2}$

b 10.0

8 a $x \left(\frac{1 - \left(\frac{2}{5}x\right)^n}{1 - \frac{2}{5}x} \right)$

b 2555

9 $\frac{1 - 4 \times \left(\frac{1}{4}\right)^n + 3 \times \left(\frac{1}{5}\right)^n}{12}$

10 a i $n = 10$

ii $S_{10} = -1364$

b i $n = 6$

ii $S_6 = \frac{1365}{16}$

11 a $n = 8$

b $n = 9$

c $n = 6$

d $n = 6$

12 $r = 3$



Infinite series

15 The absolute value of the common ratio must be less than 1.

16

a

i $r = 2$

ii Divergent

b

i $r = -4$

ii Divergent

c

i $r = \frac{1}{2}$

ii Convergent

d

i $r = \frac{1}{4}$

ii Convergent

17

a

i $r = \frac{1}{4}$

ii $\frac{8}{3}$

b

i $r = \frac{1}{5}$

ii $\frac{625}{4}$

c

i $r = -\frac{1}{2}$

ii $\frac{32}{3}$

d

i $r = -\frac{1}{4}$

ii $\frac{112}{5}$

18

a

$r = \frac{1}{2}$

b

$2, \frac{1}{2}, \frac{1}{4}$

19 $x < 8$ and $x > 6$

20 $x = \frac{1}{5}$

21 576, 144, 36

22

a $n = 11$

b 9

23

a $r = \frac{\sqrt{5}}{5}$

b $\frac{(25 + 5\sqrt{5})}{4}$

24 $\frac{1}{3}$



Sigma notation

25

a 340

b 7205

c 31

d 4914

e -252

f 242

g $\frac{3267}{4}$

h $-\frac{40}{27}$

26 $\sum_{k=1}^{11} \left(-\frac{1}{2}\right)^{k-1}$

27

a 4

b 4

c $\frac{1}{12}$

d $-\frac{7}{4}$

e 9

f $\frac{1}{8}$

Repeating decimals

28

a $\frac{8}{10} + \frac{8}{100} + \frac{8}{1000} + \frac{8}{10\,000} + \frac{8}{100\,000}$

b $a = \frac{8}{10}, r = \frac{1}{10}$

c $\frac{8}{9}$

29

a $\frac{44}{100}$

b $\frac{44}{10\,000}$

c $\frac{44}{1\,000\,000}$

d $a = \frac{44}{100}, r = \frac{1}{100}$

e $\frac{4}{9}$

30



a

i $0.6 + 0.06 + 0.006 + 0.0006 + \dots$

ii $\frac{2}{3}$

b

i $0.06 + 0.006 + 0.0006 + 0.00006 + \dots$

ii $\frac{1}{15}$

31

a $0.252525\dots$

b $\frac{25}{99}$

32 $\frac{2}{9}$

33 $\frac{19}{33}$

34 $\frac{994}{999}$

Applications

35

a 0.04

b $5000 \times 1.04 + 700$

c $5000 \times (1.04)^2 + 700 \times 1.04 + 700$

d $5000 \times (1.04)^3 + 700 \times (1.04)^2 + 700 \times 1.04 + 700$

e $5000 \times (1.04)^n + \frac{700((1.04)^n - 1)}{1.04 - 1}$

f \$13 293

36

a

i $50\,000 \times 1.059 + 3000$

ii $50\,000 \times (1.059)^2 + 3000 \times 1.059 + 3000$

iii $50\,000 \times (1.059)^3 + 3000 \times (1.059)^2 + 3000 \times 1.059 + 3000$

b $\frac{3000((1.059)^n - 1)}{0.059}$

c \$99 791



37

a $A_1 = 30\,000 \times 1.015 - 600$

b

i $A_2 = 30\,000 \times (1.015)^2 - (1.015 \times 600) - 600$

ii $A_3 = 30\,000 \times (1.015)^3 - (1.015^2 \times 600) - (1.015 \times 600) - 600$

c $A_n = 30\,000 \times (1.015)^n - \frac{600((1.015)^n - 1)}{1.015 - 1}$

d \$22\,908.60

38

a $1400 \times (1.005)^n + \frac{165((1.005)^n - 1)}{(1.005) - 1}$

b \$8165.81

39

a $A_1 = 190\,000 \times 1.07 - x$

b

i $A_2 = 190\,000 \times (1.07)^2 - 1.07x - x$

ii $A_3 = 190\,000 \times (1.07)^3 - (1.07)^2 x - 1.07x - x$

c $190\,000 \times (1.07)^n - \frac{x((1.07)^n - 1)}{0.07}$

d \$17\,934.66

40

a $110\,000 \times 1.04^n - \frac{x(1.04^n - 1)}{0.04}$

b $x = 4488.88$

41

a $190\,000 \times 1.0025^n - \frac{x(1.0025^n - 1)}{0.0025}$

b $x = 801.05$